Hospital Readmission Risk Predictor
A Phased MLOps Build

Baseline Model Fitting & Interpretation Report

Prepared for: Hospital Administration & Care Management Stakeholders

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GitHub Repository:
https://github.com/AngelineSetiawan/AngelineSetiawan-hospital-readmission-risk-predictor

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1. Background & Question

This report is the third deliverable in the Hospital Readmission Risk Predictor project, following the Exploratory Data Analysis (EDA) Report submitted on July 21, 2026, and the Preprocessing & Feature Engineering Report submitted on July 28, 2026. Where those reports characterized and prepared the data, this report documents the first fitted model: a regularized logistic regression baseline built on the leak-free, patient-level table produced in Phase 2. The research question, hypothesis, and prediction carried over from the proposal are restated below for continuity, since no changes were made to any of them during this phase.

Research Question: Can a patient's risk of being readmitted to the hospital within 30 days of discharge be predicted from routinely captured clinical encounter data, and can that prediction be delivered through a live, self-updating system rather than a one-time report?

Hypothesis & Prediction: Patients with more complex or intensive care histories, meaning longer hospital stays, more prior inpatient or emergency visits, and a higher number of medications, are more likely to be readmitted within 30 days, because these factors reflect greater underlying disease severity and more complicated post-discharge care needs. The corresponding prediction is that a supervised classification model trained on the resulting features will outperform a naive majority-class baseline on AUC-ROC and recall for the readmitted class, and that utilization-related features, specifically prior inpatient visits, time in hospital, and number of medications, will rank among the model's most important predictors.

This baseline model is the first direct test of that prediction: before any hyperparameter tuning or more flexible model family is introduced, does a simple, interpretable classifier already show discrimination above chance, and do the utilization variables identified in the EDA and Preprocessing reports carry through as important predictors once the model is actually fit?

2. Methods

2.1 Data and Split

The primary analysis dataset is the patient-level snapshot (one row per patient, first eligible encounter, excluding encounters ending in in-hospital death) created during Phase 2 feature engineering. Phase 2 produced a time-aware 75/25 train and test split with zero patient overlap: 52,829 training patients and 17,610 testing patients. The positive class (readmitted_30d = 1) makes up 9.64% of the training set and 6.82% of the test set, confirming a meaningfully imbalanced outcome.

2.2 Feature Set

Following Phase 2 feature engineering, identifiers and ordering fields (patient_nbr, encounter_id, encounter_order), the raw readmitted source field, high-cardinality raw ICD-9 diagnosis codes (replaced by the grouped diag_1_group, diag_2_group, and diag_3_group fields), and discharge-disposition fields were excluded from the baseline feature matrix. Discharge fields were dropped specifically because the intended use case is a risk score generated around admission or early care, not a prediction that relies on information only known after discharge planning. This left 49 modeling features (10 numeric, 39 categorical) before encoding.

2.3 Baseline Model Choice

Regularized logistic regression (L2 penalty, C = 1.0, liblinear solver) was selected as the baseline because it is interpretable (coefficients map directly to log-odds and odds ratios), computationally practical on roughly 53,000 training rows, and provides a transparent reference point before moving to more flexible models in later phases. The regularization strength was intentionally left at the standard default rather than tuned, since hyperparameter tuning is reserved for the next phase.

2.4 Preprocessing Pipeline

All preprocessing was placed inside a scikit-learn Pipeline so that transformation parameters are learned only from training data within each cross-validation fold, preventing preprocessing leakage. Numeric features were median-imputed and standardized; categorical features were imputed to an explicit “Missing” category and one-hot encoded, with unknown categories at test time ignored rather than raising an error.

2.5 Cross-Validation

Stratified 5-fold cross-validation was performed on the training partition only, preserving the roughly 9.6% positive-class rate in each fold. PR-AUC, ROC-AUC, precision, recall, F1, and accuracy were all scored; PR-AUC was treated as the primary metric because it is more informative than ROC-AUC or accuracy under class imbalance. The held-out Phase 2 test set (a later, time-aware slice of patients) was not touched during model selection and was reserved strictly for final evaluation.

2.6 Assumption Checks

Because logistic regression does not require normally distributed predictors or residuals, the assumptions evaluated were: (1) independence of observations, assessed via a zero patient-overlap check between train and test; (2) approximate linearity in the log-odds for continuous predictors, assessed heuristically by binning each continuous predictor into deciles and plotting empirical log-odds of readmission against the bin median; (3) absence of severe multicollinearity among numeric predictors, screened using Variance Inflation Factor (VIF); (4) no evidence of a convergence or perfect-separation problem, checked by confirming the solver converged before the iteration limit; and (5) probability calibration, examined as a supplementary diagnostic.

3. Results

3.1 Fitted Model Performance

Table 1 summarizes 5-fold cross-validated performance on the training set, and Table 2 summarizes performance on the held-out test set using a default 0.50 classification threshold.

Metric	CV Mean	CV SD
PR-AUC	0.1597	0.0038
ROC-AUC	0.6326	0.0022
Precision	0.4317	0.1004
Recall	0.0049	0.0022
F1	0.0097	0.0044
Accuracy	0.9034	0.0001

Table 1. Five-fold stratified cross-validation results (training partition).

Metric	Test Score
PR-AUC	0.1083
ROC-AUC	0.6046
Precision	0.3000
Recall	0.0025
F1	0.0050
Accuracy	0.9316

Table 2. Held-out test-set performance at a 0.50 classification threshold.

PR-AUC of 0.108 to 0.160 (versus a positive base rate of roughly 7 to 10%) indicates the model carries modest, above-baseline discriminative signal, and ROC-AUC around 0.60 to 0.63 confirms performance clearly better than chance (0.50) but far from strong separation. At the default 0.50 threshold, the confusion matrix (16,402 true negatives, 7 false positives, 1,198 false negatives, 3 true positives) shows the model almost never predicts the positive class: recall is only 0.25%. This is expected behavior for a class-imbalanced problem with an unadjusted default threshold, and it is consistent with our prediction of limited detection at 0.50; overall accuracy (93%) is misleadingly high and should not be trusted as a headline metric.

Coefficient interpretation: the strongest positive associations with 30-day readmission were medical specialties such as Hematology/Oncology (OR ≈ 2.27), Allergy and Immunology (OR ≈ 2.10), and Oncology (OR ≈ 2.08), along with an "Up" dosage-change indicator for repaglinide (OR ≈ 1.87). The strongest negative associations were Pediatrics-Endocrinology (OR ≈ 0.25), Surgery-Cardiovascular (OR ≈ 0.42), and Otolaryngology (OR ≈ 0.47). As with any one-hot-encoded logistic regression, these are associations relative to a reference category within the fitted model, not causal effects.

3.2 Assumption Diagnostics

Independence: zero patients appeared in both training and test sets, which is an important safeguard against patient-level leakage, and the one-row-per-patient design removes one obvious source of dependence between observations. This check, however, does not by itself establish full statistical independence across all observations. The dataset does not expose which hospital or provider treated each patient, so clustering at the hospital or provider level, or any other unmeasured grouping, cannot be ruled out. The independence assumption is therefore reasonably well supported, conditional on the absence of such clustering; if patients are concentrated within a small number of hospitals or providers, or share some other unobserved grouping, observations that are formally in different partitions could still be correlated.

Linearity in the log-odds: binned log-odds plots for the seven eligible continuous predictors (admission_source_id, time_in_hospital, num_lab_procedures, num_medications, number_outpatient, number_emergency, number_inpatient) showed mostly smooth, roughly monotonic trends, for example time_in_hospital and num_medications rise steadily with log-odds before leveling off, supporting approximate linearity for most predictors, though a few show mild curvature that could be revisited with polynomial terms or binning in later phases.

Table 3 reports VIF for the numeric predictors (categorical dummies excluded, since naive VIF on one-hot columns is not meaningfully interpretable).

Feature	VIF
number_diagnoses	8.33
num_medications	6.97
num_lab_procedures	5.97
time_in_hospital	4.33
admission_type_id	2.84
admission_source_id	2.83
num_procedures	2.09
number_inpatient	1.14
number_outpatient	1.09
number_emergency	1.08

Table 3. Variance Inflation Factor for numeric predictors.

All VIF values are below the 10.0 heuristic threshold for severe multicollinearity; number_diagnoses (8.33), num_medications (6.97), and num_lab_procedures (5.97) are elevated enough to note but not so high as to require immediate remediation. The solver converged in 14 of a maximum 2,000 iterations, with no indication of a separation or convergence problem. The calibration curve tracked closely to the diagonal across the observed probability range (roughly 0 to 0.20), indicating the model's predicted probabilities are reasonably well calibrated, though this range is narrow because predicted probabilities rarely exceed about 0.20.

3.3 Overfitting Check

Table 4 compares mean training and mean validation scores across the five cross-validation folds.

Metric	Mean Training Score	Mean Validation Score	Train minus Validation
PR-AUC	0.1715	0.1597	0.0119
ROC-AUC	0.6534	0.6326	0.0208
F1	0.0100	0.0097	0.0003

Table 4. Training-versus-validation performance gap.

The gaps between training and validation scores are small (PR-AUC gap of approximately 0.012, ROC-AUC gap of approximately 0.021), and test-set PR-AUC and ROC-AUC are close to their cross-validated counterparts. This indicates the L2-regularized baseline is not meaningfully overfitting the training data; overfitting was further controlled through the pipeline design (all preprocessing learned only on training folds), stratified 5-fold cross-validation, and the L2 penalty itself. Overfitting will receive a more detailed treatment, including regularization-strength tuning, next week.

4. Discussion & Next Steps

4.1 Key Takeaways

- **A modest but genuine relationship exists.** The baseline logistic regression confirms a real but modest relationship between admission-time features and 30-day readmission risk. ROC-AUC (about 0.60 to 0.63) and PR-AUC (about 0.11 to 0.16, above the roughly 7 to 10% base rate) both indicate the model is picking up genuine signal, partially supporting the hypothesis that utilization-history and clinical-encounter variables carry information about readmission risk.
- **The default threshold is not clinically usable.** At the default 0.50 threshold the model essentially never flags a patient as high-risk, so in its current form it would not be clinically useful for identifying patients who need intervention. This motivates the threshold-selection step planned for the next phase.
- **Feature importance partially revises the hypothesis.** The question of which features matter most is partially answered: medical specialty categories dominate the largest coefficients, more so than the core utilization-history variables (number_inpatient, number_emergency, number_outpatient) originally hypothesized to be most influential. This suggests the analysis plan should place more emphasis on interaction effects or non-linear treatment of specialty and utilization variables going forward, and that the original hypothesis about utilization history being the dominant driver needs to be revised in light of this baseline evidence.
- **The analysis plan is revised, not restarted.** The original plan assumed a single baseline model would be sufficient to gauge feature importance directly from coefficients. Given the imbalance-driven threshold problem observed here, the plan is revised to add an explicit threshold-selection step and to compare logistic regression against a tree-based ensemble that may capture non-linear and interaction effects the current linear specification misses, for example in num_lab_procedures and time_in_hospital, which showed mild curvature in the log-odds plots.

4.2 Planned Next Steps: Hyperparameter Tuning & Model Evaluation

- Tune the logistic-regression regularization strength (C) using training data only, via cross-validation.
- Fit and compare a tree-based ensemble (e.g., random forest or gradient boosting) as an alternative candidate model.
- Continue using PR-AUC as the primary comparison metric, with ROC-AUC and recall as secondary metrics.
- Investigate the mild non-linear relationships flagged in the log-odds diagnostic plots (e.g., binning or polynomial terms).
- Evaluate class weighting or resampling to see whether clinically useful recall can be improved without an unacceptable precision trade-off.
- Select a decision threshold using a validation set rather than the test set, since 0.50 is not appropriate for this class distribution.
- Conduct a more detailed model-evaluation and error-analysis step before selecting a final candidate model for deployment.

Works Cited

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Appendix. Data Dictionary and Codebook

The primary dataset contains one row per patient and uses the engineered readmitted_30d target. Phase 2 reported a cleaned primary patient-level dataset of 70,439 patients, followed by a 75/25 time-aware train and test split producing 52,829 training patients and 17,610 testing patients, with zero patient overlap.

A.1 Target

Variable	Description
readmitted_30d	Binary outcome: 1 if the encounter/patient snapshot corresponds to readmission within 30 days (readmitted == "<30"), otherwise 0.

A.2 Key Numeric Predictors

Variable	Description
time_in_hospital	Number of days in the hospital for the encounter.
num_lab_procedures	Number of laboratory procedures during the encounter.
num_procedures	Number of procedures during the encounter.
num_medications	Number of medications during the encounter.
number_outpatient	Number of outpatient visits in the prior year.
number_emergency	Number of emergency visits in the prior year.
number_inpatient	Number of inpatient visits in the prior year.
number_diagnoses	Number of diagnoses recorded for the encounter.

A.3 Engineered Categorical Predictors

Variable	Description
race	Patient race category, with missing values explicitly represented as Missing.
gender	Patient gender.

Variable	Description
age	Age bucket.
medical_specialty	Medical specialty, with missing values represented as an explicit category.
payer_code	Payer category, with missing values represented as an explicit category.
admission_type_id_label	Human-readable admission-type category created from the Phase 2 ID mapping.
admission_source_id_label	Human-readable admission-source category created from the Phase 2 ID mapping.
diag_1_group / diag_2_group / diag_3_group	Grouped clinical category for the primary, secondary, and tertiary ICD-9 diagnosis.

A.4 Informative Missingness Indicators

Variable	Description
max_glu_serum_tested	Indicates whether maximum glucose serum testing was performed.
A1Cresult_tested	Indicates whether an A1C test was performed.

A.5 Modeling Exclusions

Patient identifiers, encounter identifiers and ordering fields, the original readmitted target source field, raw high-cardinality diagnosis codes, and discharge-disposition fields are excluded from the primary baseline feature matrix. Excluding the original readmitted field is a direct leakage safeguard because readmitted_30d was derived from it.